Deep Learning-Based Super-Resolution of Satellite Imagery up to 0.3 m Ground Sampling Distance (GSD)

Version: 1.2

Document Type: Software Requirements Specification (SRS)

Project Duration: 8–12 Months

Target Users: Final Year Undergraduate Students (Computer Science, Artificial Intelligence, Remote Sensing, GIS, Software Engineering)

1. Introduction

1.1 Background

Very High Resolution (VHR) satellite imagery is essential for applications such as urban planning, military reconnaissance, disaster monitoring, agriculture, environmental analysis, and autonomous mapping. However, imagery acquired by satellites is constrained by sensor optics, detector size, orbit altitude, atmospheric effects, and imaging physics. Consequently, acquiring imagery at extremely fine spatial resolutions (e.g., 0.3 m GSD) is expensive and not always possible.

Artificial intelligence based Super-Resolution (SR) aims to reconstruct a plausible high-resolution image from one or more lower-resolution observations by learning mappings from paired datasets. Unlike traditional interpolation methods, SR attempts to recover missing high-frequency image content while maintaining geometric and radiometric consistency.

1.2 Problem Statement

Current interpolation techniques (Nearest Neighbor, Bilinear, Bicubic, Lanczos etc..) merely increase the number of pixels and do not recover lost spatial information. Such methods produce visually larger images but do not improve true spatial resolution.

The project aims to develop an artificial intelligence based SR framework that reconstructs high-resolution satellite imagery while preserving geometric structures, radiometric integrity, and geospatial metadata. The goal is to approach 0.3 m GSD from 0.5, 1, and 2.5m GSD.

1.3 Objectives

Design an end-to-end satellite image super-resolution framework.

- Support GeoTIFF and remote sensing datasets.

- Preserve geospatial metadata.
- Generate super-resolved imagery using deep learning.
- Compare multiple SR architectures.
- Quantitatively evaluate reconstruction quality.
- Assess downstream performance (e.g., object detection).
- Produce reproducible experimental results.

2. Scope

The project covers:

- Single-image super-resolution (primary objective)
- AI learning model training
- Dataset preparation
- Accuracy assessment
- Visualization
- GeoTIFF export
- Research comparison The project does not include:
    - Image compression
    - Pan-sharpening (unless studied separately)
    - Image interpolation as the final solution

3. Functional Requirements

FR-1 Data Import and Validation

The system shall import satellite imagery in standard geospatial formats (e.g., GeoTIFF *mandatory*, other formats *optional*) and also verify file integrity, spatial reference information, bit depth, band configuration, and image dimensions before processing.

FR-2 Geospatial Metadata Preservation

The system shall preserve all essential geospatial metadata, including Coordinate Reference System (CRS), affine transformation, pixel size, image extent, and projection information throughout the processing workflow and in the final output.

FR-3 Data Preprocessing

The system shall preprocess input imagery to prepare datasets suitable for model training and evaluation by performing radiometric normalization, optional contrast enhancement, image registration (if required), patch generation, and data augmentation.

FR-4 Training Dataset Generation

The system shall generate paired low-resolution and high-resolution image patches, organize training, validation, and testing datasets, and support configurable scaling factors and patch sizes.

FR-5 Model Training

The system shall train deep learning super-resolution models using configurable hyper parameters, save model checkpoints, monitor training progress, and record performance metrics for reproducible experiments.

FR-6 Model Inference

The system shall accept low-resolution satellite imagery as input and generate super-resolved imagery while maintaining geometric alignment and preserving geospatial metadata.

FR-7 Accuracy Assessment

The system shall automatically evaluate generated outputs using objective image quality, spectral, geometric, and perceptual assessment metrics, and generate comprehensive performance reports.

FR-8 Visualization and Comparison

The system shall provide visual comparison of input, reference, and super-resolved images through synchronized display, zooming, difference maps, and quality metric summaries to support qualitative analysis.

FR-9 Batch Processing

The system shall support automated processing of multiple images or datasets in a single execution, enabling efficient large-scale experimentation and evaluation.

FR-10 Output Generation

The system shall export super-resolved images in geospatial formats while preserving metadata and generate accompanying reports containing processing parameters, evaluation metrics, and experimental results.

FR-11 Experiment Management

The system shall maintain logs of training configurations, model versions, datasets, processing parameters, and evaluation results to ensure experiment traceability and reproducibility.

FR-12 User Configuration

The system shall provide configurable parameters for dataset preparation, model selection, training settings, inference options, and evaluation metrics without requiring modification of the source code.

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4. Supported Data

4.1 Satellite Platforms

- Maxar WorldView
- GeoEye
- Pleiades
- Sentinel-2 (research)
- SPOT
- PlanetScope

4.2 Supported Bit Depth

- 8-bit
- 10-bit
- 12-bit

4.3 Spectral Support

- RGB
- Panchromatic
- Multispectral

5. Resolution Accuracy Requirements

- RA-1: The software shall not claim that interpolation methods increase true spatial resolution.
- RA-2: The software shall clearly distinguish between:
    - Image enlargement
    - Image enhancement
    - Image restoration
    - Super-resolution
    - Pan-sharpening
- RA-3: The software shall avoid generating unrealistic textures and artifacts that could mislead interpretation.
- RA-4: Edges of buildings, roads, water bodies, vegetation, and other objects shall remain geometrically consistent.
- RA-5: The software shall preserve:

    - Spectral consistency
    - Radiometric consistency
    - Geometric consistency
    - Geospatial metadata
- RA-6: The software shall clearly state the uncertainty of generated details and clarify that reconstructed information is based on model inference, not guaranteed to be accurate.

6. Quantitative Success Criteria

The project shall be considered successful if it satisfies the following quantitative requirements on an independent test dataset.

6.1 Accuracy Assessment

The proposed software will be evaluated against bicubic interpolation as the baseline method. The same input images will be processed using both methods, and their results will be compared.

PSNR: The proposed method should improve PSNR by at least 1.5 dB over the bicubic baseline.

SSIM: The proposed method should achieve an SSIM of at least 0.90 or improve SSIM by at least 0.03 over the bicubic baseline.

RMSE: The proposed method should reduce RMSE by at least 10% compared with the bicubic baseline.

Geometric Registration: The mean image registration error should be ≤ 0.5 pixels.

Metadata Preservation: The software should preserve 100% of the CRS, affine transform, pixel size, and georeferencing metadata.

Inference Time: The software should process a 512 × 512 pixel image patch in ≤ 5 second on the target GPU.

Visual Quality: The output should have no visually significant checkerboard artifacts, ringing, or excessive/unrealistic texture generation when compared with the baseline and evaluated visually.

Baseline: Bicubic interpolation is used as the baseline because it is a standard non-AI image upscaling/interpolation method. The proposed method should demonstrate measurable improvements over this baseline.

6.2 Validation Criteria

The generated super-resolved imagery shall demonstrate:

- Preservation of building boundaries.
- Preservation of road centerlines and widths.
- Preservation of vegetation boundaries.

- No systematic geometric distortion.
- No noticeable spectral shifts.
- Minimal artificial textures or hallucinated objects.
- Consistent radiometric characteristics with the reference imagery.

6.3 Acceptance Criteria

The system will be accepted if:

- At least 80% of the quantitative performance criteria are achieved.
- The super-resolution model outperforms the interpolation baseline in PSNR, SSIM, and at least one perceptual quality metric.
- Geospatial metadata remains fully preserved in the output products.
- Independent visual inspection by project supervisors confirms that the reconstructed imagery is free from obvious artifacts that could mislead interpretation.
- Experimental results are reproducible using the documented code, datasets, and trained model.